

Fuzzy Set Practical - Important Terms, Definitions and Formulae

Fuzzy Set

Definition	A fuzzy set is a set in which elements can partially belong with membership values between 0 and 1.
Formula	$0 \leq \mu(x) \leq 1$

Membership Function

Definition	A membership function defines the degree of belongingness of an element in a fuzzy set.
Formula	$\mu(x)$

Union of Fuzzy Sets

Definition	Union combines two fuzzy sets using maximum membership values.
Formula	$\mu(A \cup B)(x) = \max(\mu A(x), \mu B(x))$

Intersection of Fuzzy Sets

Definition	Intersection combines fuzzy sets using minimum membership values.
Formula	$\mu(A \cap B)(x) = \min(\mu A(x), \mu B(x))$

Complement of Fuzzy Set

Definition	Complement represents opposite membership of a fuzzy set.
Formula	$\mu(A^c)(x) = 1 - \mu A(x)$

Difference of Fuzzy Sets

Definition	Difference represents elements belonging to A but not to B.
Formula	$\mu(A-B)(x) = \min(\mu A(x), 1 - \mu B(x))$

Fuzzy Relation

Definition	A fuzzy relation represents relationship between elements of two fuzzy sets.
Formula	$R \subseteq A \times B$

Cartesian Product

Definition	Cartesian product creates all possible ordered pairs between two fuzzy sets.
Formula	$\mu(A \times B)(x,y) = \min(\mu A(x), \mu B(y))$

Max-Min Composition

Definition	Max-Min composition combines two fuzzy relations using minimum followed by maximum operation.
Formula	$R \circ S = \max(\min(R,S))$

Ordered Pair

Definition	An ordered pair contains two elements written in fixed order.
Formula	$(a,b) \neq (b,a)$

Degree of Membership

Definition	Degree of membership indicates how strongly an element belongs to a fuzzy set.
Formula	$0 \leq \mu(x) \leq 1$

Crisp Set

Definition	A crisp set is a traditional set where an element either completely belongs or does not belong.
Formula	Membership = 0 or 1